competishun
Power of real gurus

MATHEMATICS

TARGET : JEE-ADVANCED-2024

CAPS-6^{+}

SCQ (Single Correct Type) :

1. If f(x) = $[x^2] + \sqrt{\{x\}^2}$, where [.] and {.} denote the greatest integer and fractional part functions respectively, then-

(A) f(x) is continuous at all integral points except 0

(B) f(x) is continuous and differentiable at x = 0

(C) f(x) is discontinuous for all $x \in I - \{1\}$

(D) f(x) is not differentiable for all $x \in I$

2. The function $f(x) = (x^2 - 1)\,|\,x^2 - 3x + 2\,| + \cos(|\,x\,|)$ is not differentiable at

(A) –1 (B) 0 (C) 1 (D) 2.

3. Let $f(x) = \begin{cases} \dfrac{\frac{1}{e^{x-2}} - 1}{\frac{1}{3^{x-2}} + 1} & x > 2 \\ \dfrac{b\sin\{-x\}}{\{-x\}} & x < 2 \\ c & x = 2 \end{cases}$, where {·} denotes fraction part function, is continuous at x = 2, x = 2

then b + c =

(A) 0 (B) 1 (C) 2 (D) 4

4. The point on the curve $4x^2 + a^2y^2 = 4a^2$, $a^2 \in (4, 8)$. Which is farthest from P(0, –2) is

(A) (0, 2) (B) (a, 0) (C) (–4, 0) (D) None of these

5. if $y = x^3 - c$ and $y = x^2 + ax + b$ touch each other at the point (1, 2) then $\dfrac{|a|+|b|+|c|}{|a+b-c|}$

(A) 1 (B) 0 (C) 2 (D) none of these

6. A person is standing at the edge of a slow moving river which is 1 km wide. He wishes to return to the camp ground on the opposite side of the river for which he may swim to any point on the opposite bank and then walk for the rest of the distance. The campground is 1 km away from the point on the opposite bank directly across from where he starts to swim. If he swims at the rate of 2 km/hr and walks at the rate of 3 km/hr, the minimum time taken by him is approximately

(A) 0.6 hr (B) 0.7 hr (C) 0.8 hr (D) 0.9 hr

7. If f(x) is a thrice differentiable function such that, $\lim\limits_{x\to 0}\dfrac{f(4x)-3f(3x)+3f(2x)-f(x)}{x^3}=12$ then the value of $f'''(0)$ equals to :

(A) 0 (B) 1 (C) 12 (D) None of these

8. $$y=\frac{1}{1+(\tan\theta)^{\sin\theta-\cos\theta}+(\cot\theta)^{\cos\theta-\cot\theta}}+\frac{1}{1+(\tan\theta)^{\cos\theta-\sin\theta}+(\cot\theta)^{\sin\theta-\cot\theta}}$$

$+\dfrac{1}{1+(\tan\theta)^{\cos\theta-\cot\theta}+(\cot\theta)^{\cot\theta-\sin\theta}}$ then $\dfrac{dy}{dx}$ at $\theta=\dfrac{\pi}{3}$ is :

(A) 0 (B) 1 (C) $\sqrt{3}$ (D) None of these

MCQ (One or more than one correct) :

9. If f : R $\to$ R be a differentiable function and f(0) = 0 and f′(0) = 1, then

$\ell\text{im}_{x\to 0}\dfrac{1}{x}\left[f(x)+f\left(\dfrac{x}{2}\right)+f\left(\dfrac{x}{3}\right)+\ldots+f\left(\dfrac{x}{n}\right)\right]$, where n $\in$ N, equals

(A) 0

(B) $1+\dfrac{1}{2}+\dfrac{1}{3}+\ldots.+\dfrac{1}{n}$

(C) ${}^nC_1-\dfrac{{}^nC_2}{2}+\dfrac{{}^nC_3}{3}-\ldots\ldots\ldots+(-1)^{n-1}\dfrac{{}^nC_n}{n}$

(D) does not exist

10. Let $f(x)=\begin{cases}\dfrac{a(1-x\sin x)+5+(b\cos x)}{x^2} & ; \quad x<0\\ 3 & ; \quad x=0\\ \left(1+\left(\dfrac{cx+dx^3}{x^2}\right)\right)^{\frac{1}{x}} & ; \quad x>0\end{cases}$ If f is continuous at x = 0 then correct statement(s) is/are :

(A) a + c = –1

(B) b + c = –4

(C) a + b = –5

(D) c + d = an irrational number

11. Let f : $R^+ \to$ R defined as f(x) = e^x + ln x and g = f^{-1} then correct statement(s) is/are :

(A) $g''(e)=\dfrac{1-e}{(1+e)^3}$ (B) $g''(e)=\dfrac{e-1}{(1+e)^3}$ (C) $g'(e)=e+1$ (D) $g'(e)=\dfrac{1}{e+1}$

12. Let $x^{2/3} + y^{2/3} = a^{2/3}$ be a equation of curve, then which of the following is correct?

(A) Equation of tangent at $P(x_1, y_1)$ to the curve is $\frac{x}{x_1^{1/3}} + \frac{y}{y_1^{1/3}} = a^{2/3}$

(B) Length of portion of tangent intercepted between coordinate axes is constant

(C) Equation of normal at $P(x_1, y_1)$ to the curve is $xx_1^{1/3} + yy_1^{1/3} = x_1^{4/3} + y_1^{4/3}$

(D) If tangent at (x_1, y_1) meet the coordinate axes in A and B, then locus of mid point of AB is $x^2 + y^2 = \frac{a^2}{4}$

13. Tangent at point P_1(other than origin) on the curve $y = x^3$ meets the curve again at P_2. The tangent at P_2 meets the curve at P_3 and so on. Then

(A) abscissae of P_1, P_2, P_3 ...... P_n form a G.P.

(B) ordinates of P_1, P_2, P_3 ... P_n form a G.P

(C) ratio of area of $\Delta P_1P_2P_3$ and $\Delta P_2P_3P_4$ is $\frac{1}{16}$

(D) ratio of area of $\Delta P_1P_2P_3$ and $\Delta P_2P_3P_4$ is 16

14. Possible value(s) of 'a' for which curves $c_1;\ y = \sqrt{3} + \sin x,\ x \in (0, 2\pi)$ and $c_2;\ y = \frac{|x|}{2} + a$, have a common tangent, is /are

(A) $\frac{3\sqrt{3}}{2} - \frac{\pi}{6}$ (B) $\frac{3\sqrt{3}}{2} + \frac{\pi}{6}$ (C) $\frac{\sqrt{3}}{2} + \frac{5\pi}{6}$ (D) $\frac{\sqrt{3}}{2} - \frac{5\pi}{6}$

15. If $y = e^{x\sin(x^3)} + (\tan x)^x$ then $\frac{dy}{dx}$ may be equal to :

(A) $e^{x\sin(x^3)}$ $[3x^3 \cos(x^3) + \sin(x^3)] + (\tan x)^x [\ln \tan x + 2x \operatorname{cosec} 2x]$

(B) $e^{x\sin(x^3)}$ $[x^3 \cos(x^3) + \sin(x^3)] + (\tan x)^x [\ln \tan x + 2x \operatorname{cosec} 2x]$

(C) $e^{x\sin(x^3)}$ $[x^3 \sin(x^3) + \cos(x^3)] + (\tan x)^x [\ln \tan x + 2 \operatorname{cosec} 2x]$

(D) $e^{x\sin(x^3)}$ $[3x^3 \cos(x^3) + \sin(x^3)] + (\tan x)^x \left[\ln \tan x + \frac{x\sec^2 x}{\tan x}\right]$

16. f(x) is differentiable function satisfying the relationship $f^2(x) + f^2(y) + 2(xy - 1) = f^2(x + y)\ \forall\ x, y \in R$

Also $f(x) > 0\ \forall\ x \in R$ and $f(\sqrt{2}) = 2$. Then which of the following statement(s) is/are correct about f(x)?

(A) $[f(3)] = 3$ ([·] denotes greatest integer function)

(B) $f(\sqrt{7}) = 3$

(C) f(x) is even

(D) $f'(0) = 0$

Numerical Based Questions :

17. Let $f(x) = x^2 + ax + 3$ and $g(x) = x + b$, where $F(x) = \lim\limits_{n\to\infty} \dfrac{f(x) + (x^2)^n g(x)}{1 + (x^2)^n}$. If F(x) is continuous at x = 1 and x = –1 then find the value of $(a^2 + b^2)$.

18. Let f(x) be a function of real variable, so that slope of the line through any two points on the curve y = f(x) is non negative. $x = \alpha - f(\alpha)$ is a solution of the equation $x = \alpha - f[x + f(\alpha)]$ for $\alpha \in R$, then find the number of other solution(s) of the equation.

Subjective Type Problems :

19. If $f(x) = x^2 - 2|x|$, then test the derivability of g(x) in the interval [–2, 3], where

$$g(x) = \begin{cases} \min.\{f(t); -2 \le t \le x\}, & -2 \le x < 0 \\ \max.\{f(t); 0 \le t \le x\}, & 0 \le x \le 3 \end{cases}$$

20. Discuss the continuity and differentiability of the function $f(x) = \left\{\lim\limits_{n\to\infty}\left(\lim\limits_{m\to\infty} \dfrac{\cos^{2n}(m!\pi x) - 1}{\cos^{2n}(m!\pi x) + 1}\right)\right\}$

21. Find the value of f(0) so that the function

$$f(x) = \frac{\cos^{-1}(1 - \{x\}^2)\sin^{-1}(1 - \{x\})}{\{x\} - \{x\}^3}, \quad x \neq 0$$

({x} denotes fractional part of x) becomes continuous at x = 0

****

ANSWERKEY OF CAPS-6+

1.	(A)	**2.**	(D)	**3.**	(A)	**4.**	(A)	**5.**	(A)	**6.**	(B)	**7.**	(C)
8.	(A)	**9.**	(BC)	**10.**	(ABCD)	**11.**	(AD)	**12.**	(ABD)	**13.**	(ABC)	**14.**	(AD)
15.	(AD)	**16.**	(ABCD)	**17.**	17	**18.**	0						

19. discontinuous at x = 0 and not differentiable at x = 0, 2

20. discontinuous and non-differentiable

21. no value of f(0)